

LEARN PYTHON PROGRAMMING – BEGINNER TO MASTER

Section: 5. Conditional Statements

Lecture: 39. Challenge - Odd Or Even

Section 1: Lecture Summary

This lecture focuses on practical challenges using conditional statements in Python. It revisits the basic concept of conditionals and applies them to two simple but foundational problems: determining if a number is odd or even, and checking if a person is eligible to vote based on age. The instructor explains the logic behind even/odd numbers using division and remainder, introduces the modulus operator (%) in Python to get the remainder, and demonstrates how to structure if-else statements to print the appropriate results. Then, the concept of voting eligibility is introduced with the criterion of age being 18 or above. The lecture concludes with sample Python code snippets and encourages students to write and run these programs themselves for practice.

Section 2: Key Concepts and Explanations

- Conditional Statements in Python:
 - A fundamental programming construct that allows executing code based on conditions.
 - The structure: `if condition: <code> else: <code>`
- Even and Odd Numbers:
 - Even Number: An integer that is exactly divisible by 2, meaning the remainder when divided by 2 is zero.
 - Odd Number: An integer where the remainder is 1 when divided by 2.
 - The modulus operator ``%`` returns the remainder of division.
 - In Python, checking ``n % 2 == 0`` determines if ``n`` is even.

- Modulus Operator (%) in Python:
 - Given `n % 2`, it provides the remainder when `n` is divided by 2.
 - Results always 0 or 1 for division by 2.
- Input and Type Casting:
 - The `input()` function takes user input as a string.
 - Convert input to integer using `int()` for numeric operations.
- Voting Eligibility:
 - Eligibility criterion: Age ≥ 18 .
 - Use an if-else statement to determine eligibility.
 - Print "Eligible" if condition is true; otherwise, print "Not Eligible".

Section 3: Example Code and Use Cases

Program to check if a number is even or odd

```
n = int(input("Enter a number: ")) # Takes input and converts to integer

if n % 2 == 0:
    print("Even")
else:
    print("Odd")
```

This code captures user input, checks divisibility by 2 using modulus operator, and prints whether the number is "Even" or "Odd".

Program to check if a person is eligible to vote

```
age = int(input("Enter your age: ")) # Takes input and converts to integer
```

```
if age >= 18:  
    print("Eligible")  
else:  
    print("Not Eligible")
```

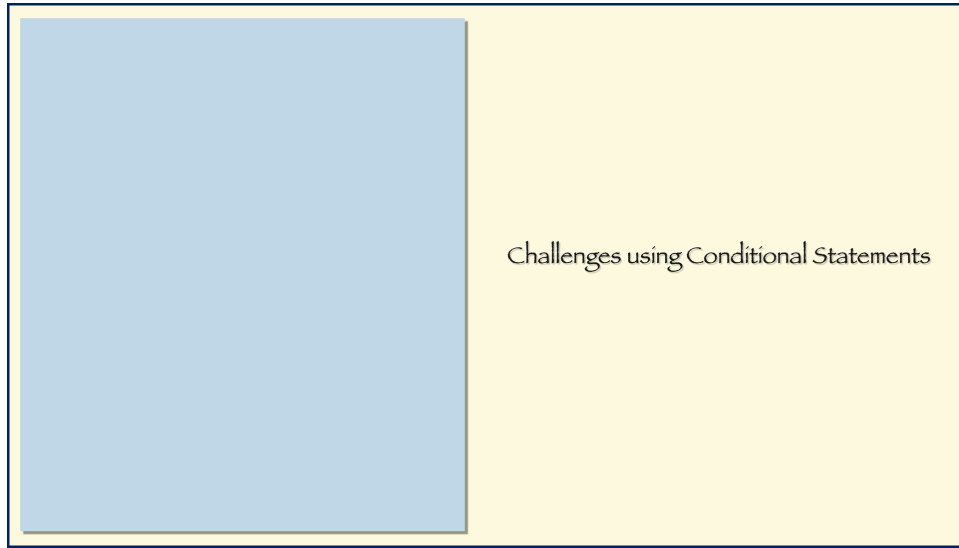
This program takes the age of a person and uses a conditional statement to check if they are eligible to vote.

Section 4: Key Takeaways

- Conditional statements enable decision making in programs based on boolean conditions.
- Even and odd numbers can be determined through the remainder after dividing by 2 using the modulus (%) operator.
- Always convert input strings to integers when performing numeric comparisons or arithmetic.
- Voting eligibility is an example of a simple conditional check using comparison operators.
- Writing simple if-else programs helps solidify understanding of logic and syntax in Python.
- Practice by typing and running these programs in an IDE like PyCharm, VS Code, or Jupyter Notebook to reinforce learning.

Lecture in Slides

Please Note: This document presents a curated selection of slides from the lecture; others have been excluded for conciseness.



Challenges using Conditional Statements

- Odd or Even

Challenges using Conditional Statements

- Odd or Even
- Eligible to Vote

Odd or Even
Challenge

Odd or Even
Challenge

n = 6

Odd or Even

Challenge

n = 6

Odd or Even

Challenge

n = 6

2) 6 (

Challenge

n = 6

$$\begin{array}{r} 2) \ 6 \ (3 \\ \underline{-6} \end{array}$$

Challenge

n = 6

$$\begin{array}{r} 2 \overline{) 63} \\ \underline{-6} \\ 0 \end{array}$$

Remainder

Odd or Even
Challenge

n = 6

2) 6 (3
-6
0
Remainder

Odd or Even
Challenge

n = 6

2) 6 (3
-6
0
Remainder

Even

Challenge

n = 9

Remainder

Even

Challenge

n = 9

-6

Remainder

Even

2) 9 (

-6

Remainder

Even

Odd or Even
Challenge

n = 6

2) 6 (3
-6
0
Remainder

Even

n = 9

2) 9 (4
-8

Odd or Even
Challenge

n = 6

2) 6 (3
-6
0
Remainder

Even

n = 9

2) 9 (4
-8
1
Remainder

Odd or Even

Challenge

n = 6

2) 6 (3

-6

0

Remainder

Even

n = 9

2) 9 (4

-8

1

Remainder

Odd or Even

Challenge

n = 6

2) 6 (3

-6

0

Remainder

Even

n = 9

2) 9 (4

-8

1

Remainder

Odd

Odd or Even

Challenge

n = 6

0

Remainder

Even

n = 9

1

Remainder

Odd

Odd or Even

Challenge

n = 6

0

Remainder

Even

n = 9

1

Remainder

Odd

n % 2

Odd or Even

Challenge

n = 6

0

Remainder

Even

n = 9

1

Remainder

Odd

n % 2 == 0

Odd or Even

Challenge

n = 6

0

Remainder

Even

n = 9

1

Remainder

Odd

if n % 2 == 0 :

Odd or Even

Challenge

n = 6

0

Remainder

Even

n = 9

1

Remainder

Odd

if n % 2 == 0 :

print('Even')

Odd or Even

Challenge

n = 6

0

Remainder

Even

n = 9

1

Remainder

Odd

if n % 2 == 0 :

print('Even')

else:

Odd or Even

Challenge

n = 6

0

Remainder

Even

n = 9

1

Remainder

Odd

if n % 2 == 0 :

print('Even')

else:

print('Odd')

Odd or Even

Challenge

```
if n % 2 == 0 :  
    print( 'Even' )  
else:  
    print( 'Odd' )
```

Odd or Even

Challenge

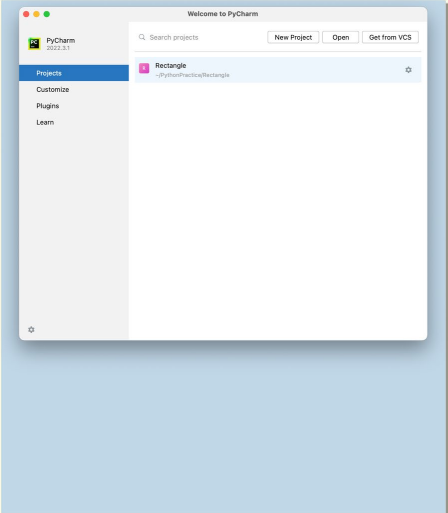
```
if n % 2 == 0 :  
    print( 'Even' )  
else:  
    print( 'Odd' )
```

Odd or Even

Challenge

```
n = int( input( 'Enter a number' ) )

if n % 2 == 0 :
    print( 'Even' )
else:
    print( 'Odd' )
```

The image shows the PyCharm 2023.3.1 Welcome screen. On the left is a sidebar with links for Projects, Customize, Plugins, and Learn. The main area is titled 'Welcome to PyCharm' and contains a search bar for projects, buttons for 'New Project', 'Open', and 'Get from VCS', and a list of recent projects with one project named 'Rectangle' visible.

Odd or Even

Challenge

```
n = int( input( 'Enter a number' ) )

if n % 2 == 0 :
    print( 'Even' )
else:
    print( 'Odd' )
```

New Project

Location: /Users/abdulbari/PythonPractice/PythonProject

Python Interpreter: New Virtualenv environment

New environment using Virtualenv

Location: /Users/abdulbari/PythonPractice/PythonProject/venv

Base interpreter: /usr/local/bin/python3.13

☐ Inherit global site-packages

☐ Make available to all projects

☐ Previously configured interpreter

Interpreter: Python 3.13 /usr/local/bin/python3.13

☒ Create a main.py welcome script

Create a Python script that provides an entry point to loading in PyCharm.

Cancel

Create

Odd or Even

Challenge

```
n = int( input( 'Enter a number' ) )

if n % 2 == 0 :
    print( 'Even' )
else:
    print( 'Odd' )
```

New Project

Location: /Users/abdulbari/PythonPractice/OddEven

Python Interpreter: New Virtualenv environment

New environment using Virtualenv

Location: /Users/abdulbari/PythonPractice/OddEven/venv

Base interpreter: /usr/local/bin/python3.13

☐ Inherit global site-packages

☐ Make available to all projects

☐ Previously configured interpreter

Interpreter: Python 3.13 /usr/local/bin/python3.13

☒ Create a main.py welcome script

Create a Python script that provides an entry point to loading in PyCharm.

Cancel

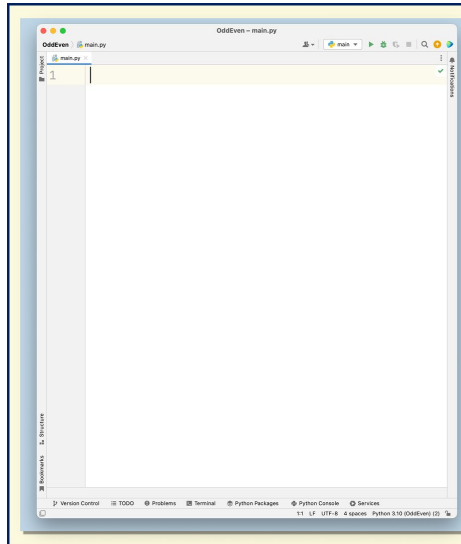
Create

Odd or Even

Challenge

```
n = int( input( 'Enter a number' ) )

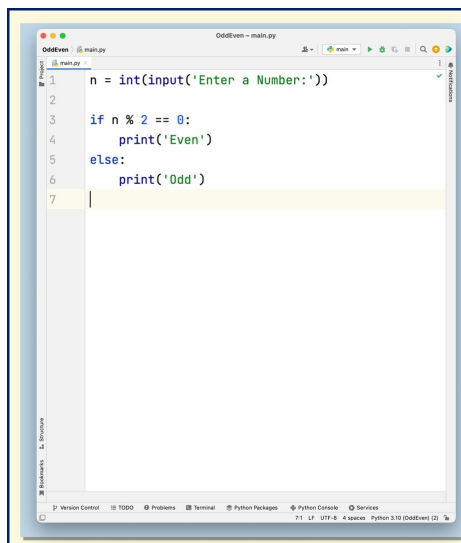
if n % 2 == 0 :
    print( 'Even' )
else:
    print( 'Odd' )
```



Odd or Even Challenge

```
n = int( input( 'Enter a number' ) )

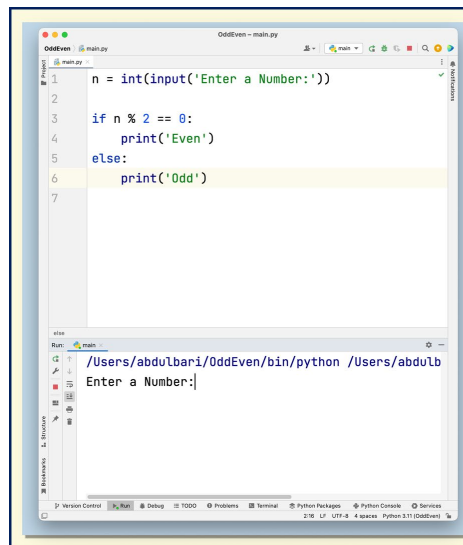
if n % 2 == 0 :
    print( 'Even' )
else:
    print( 'Odd' )
```



Odd or Even Challenge

```
n = int( input( 'Enter a number' ) )

if n % 2 == 0 :
    print( 'Even' )
else:
    print( 'Odd' )
```



The screenshot shows a code editor window titled 'OddEven - main.py'. The code is as follows:

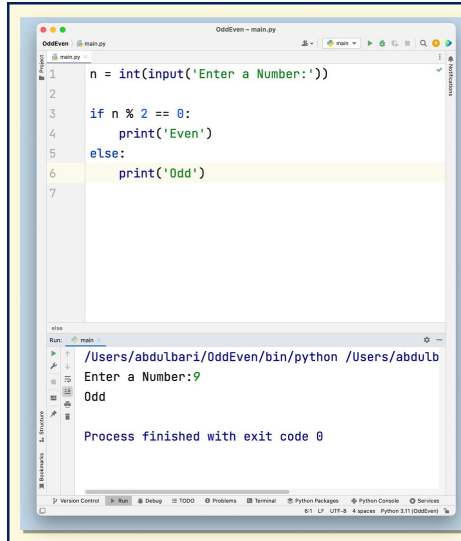
```
1 n = int(input('Enter a Number:'))
2
3 if n % 2 == 0:
4     print('Even')
5 else:
6     print('Odd')
```

Below the code editor is a 'Run' console window. It shows the command `/Users/abdu1bari/OddEven/bin/python /Users/abdu1b` and the prompt `Enter a Number:|`.

Odd or Even Challenge

```
n = int(input('Enter a number'))

if n % 2 == 0:
    print('Even')
else:
    print('Odd')
```



The screenshot shows a Python IDE window titled 'OddEven - main.py'. The code in the editor is as follows:

```
1 n = int(input('Enter a Number:'))
2
3 if n % 2 == 0:
4     print('Even')
5 else:
6     print('Odd')
7
```

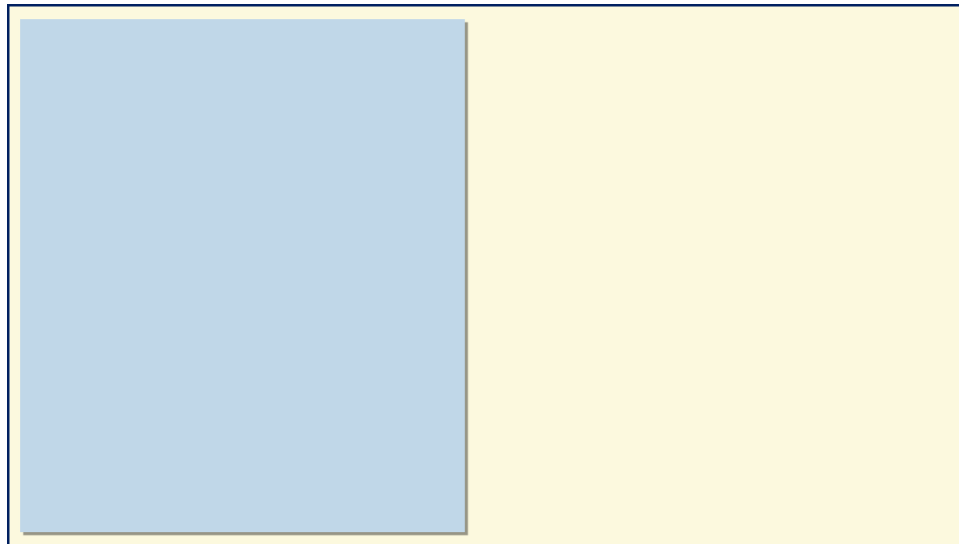
Below the editor, the 'Run' console shows the execution output:

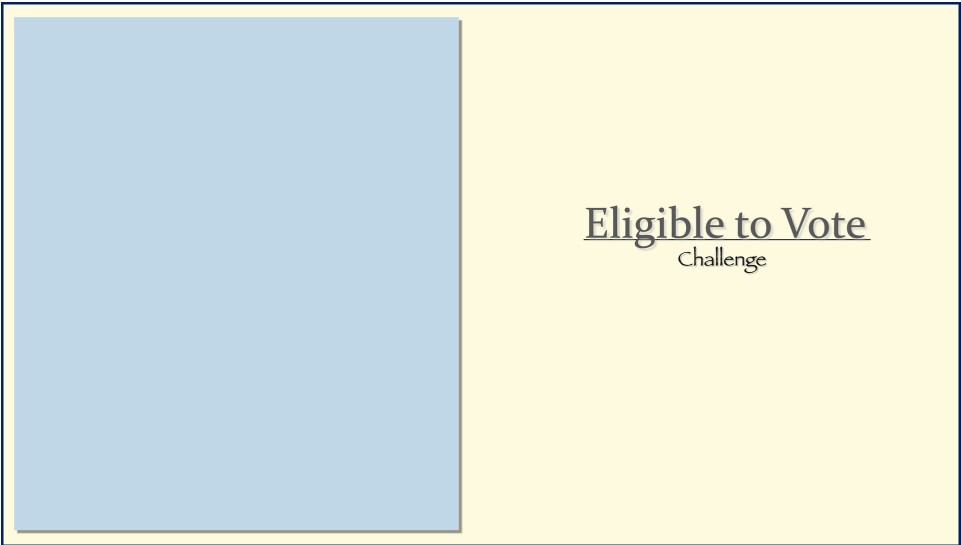
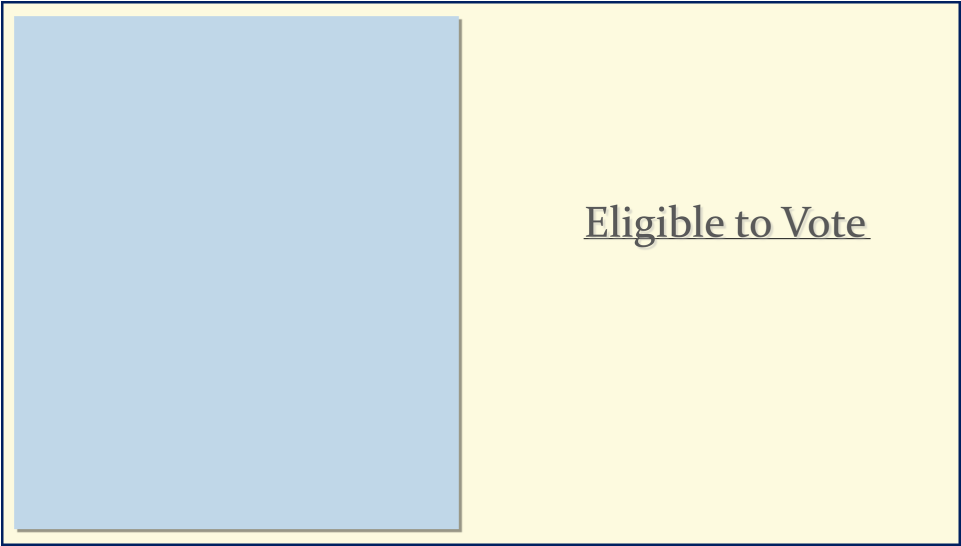
```
/Users/abdulbari/OddEven/bin/python /Users/abdulb
Enter a Number:9
Odd
Process finished with exit code 0
```

Odd or Even Challenge

```
n = int(input('Enter a number'))

if n % 2 == 0:
    print('Even')
else:
    print('Odd')
```





Eligible to Vote
Challenge

Eligible to Vote
Challenge



age = ?

Eligible to Vote

Challenge



age = ?

age >= 18

Eligible to Vote

Challenge



age = ?

age >= 18

Eligible

Eligible to Vote

Challenge



age = ?

age >= 18 Eligible

age < 18

Eligible to Vote

Challenge



age = ?

age >= 18 Eligible

age < 18 Not Eligible

Eligible to Vote

Challenge



age = ?

age = ?

Eligible to Vote

Challenge



age = ?

```
age = ?
```

```
if age >= 18 :
```

Eligible to Vote

Challenge



age = ?

```
age = ?
```

```
if age >= 18 :
```

```
    print( 'Eligible' )
```

Eligible to Vote Challenge



age = ?

age = ?

if age >= 18 :

 print('Eligible')

else:

Eligible to Vote Challenge



age = ?

age = ?

if age >= 18 :

 print('Eligible')

else:

 print('Not Eligible')